

RECRUITOS — MASTER PRODUCT REQUIREMENT DOCUMENT (PRD)

Document Status: V2.0 Engineering Final

System Architecture: Multi-Tenant Web Application (Next.js 14, PostgreSQL, Prisma/Kysely, Tailwind CSS)

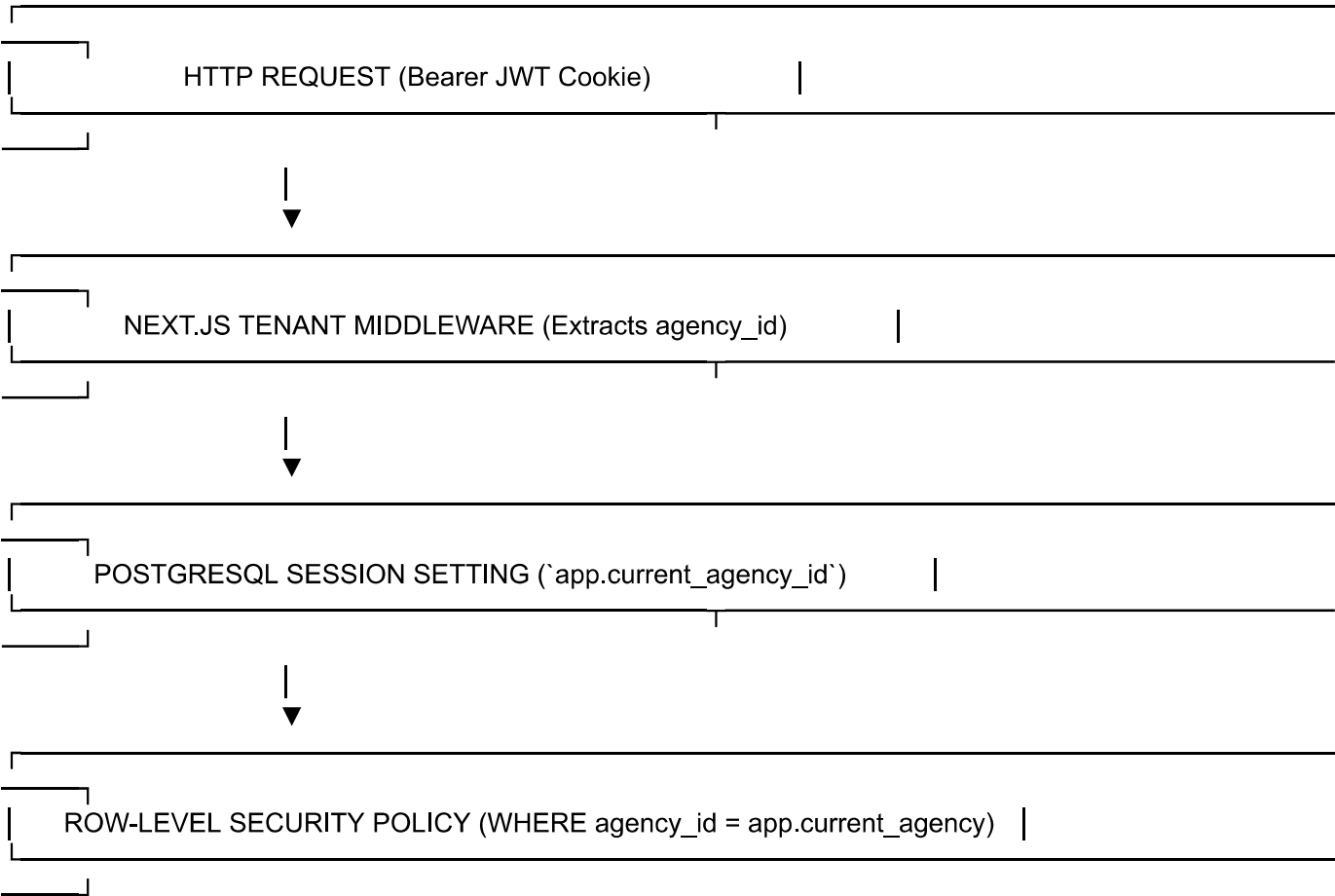
Target Builder: Divyanshu Kumar (Full-Stack Engineering Lead)

Primary Color Palette: Deep Navy (#0F172A), Botspring Yellow (#FFD400), Slate Grey (#64748B), Success Green (#10B981), Alert Red (#EF4444).

FOUNDATIONAL ARCHITECTURE: MULTI-TENANT SECURITY & TENANT ISOLATION (FOUNDATION-01)

1. Architectural Strategy

RecruitOS operates as a single-database, multi-tenant architecture where every record is scoped to an `agency_id`. Divyanshu must enforce Row-Level Security (RLS) and JWT middleware validation across every route to guarantee zero data leakage between competing recruitment agencies.



2. Executable Schema & RLS Specification

```
SQL
-- Enable Row-Level Security on Core Entities
CREATE TABLE agencies (
  agency_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  company_name VARCHAR(255) NOT NULL,
```

```

    subdomain VARCHAR(100) UNIQUE NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE users (
    user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    agency_id UUID NOT NULL REFERENCES agencies(agency_id) ON DELETE CASCADE,
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    full_name VARCHAR(255) NOT NULL,
    role VARCHAR(50) NOT NULL CHECK (role IN ('AGENCY_OWNER', 'TEAM_LEAD', 'RECRUITER')),
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Enable RLS Example on Candidates Table
ALTER TABLE candidate_records ENABLE ROW LEVEL SECURITY;

CREATE POLICY tenant_isolation_policy ON candidate_records
    AS RESTRICTIVE
    USING (agency_id = NULLIF(current_setting('app.current_agency_id', true), '')::UUID);

```

3. Middleware Security Rule

Every API controller must execute the following session scope before running database queries:

```

JavaScript
// Next.js API Context Middleware
export async function withTenantContext(req, res, handler) {
    const token = req.cookies.get('recruitos_jwt');
    const payload = verifyJwt(token);

    if (!payload || !payload.agency_id) {
        return res.status(401).json({ error: "Unauthorized: Invalid or Missing Tenant Context" });
    }

    // Inject tenant session variable for RLS
    await db.query(`SET LOCAL app.current_agency_id = '${payload.agency_id}'`);
    req.user = payload; // { user_id, agency_id, role }
    return handler(req, res);
}

```

WORKFLOW 1: DEMAND ACQUISITION & MANDATE PUBLISHING

1. Workflow Business Purpose

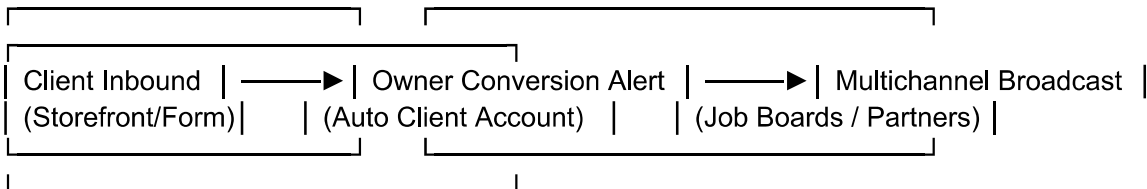
Captures inbound client hiring needs, establishes job mandates, enforces role ownership, and broadcasts positions across job portals and external partner networks while strictly protecting client identity.

2. Integrated Feature Map

- **AS-01:** Public Agency Storefront & Branded Engine
- **AS-02:** Self-Serve Client Mandate Ingestion Engine
- **RC-03:** Pipeline SLA & Stagnation Aging Radar (Mandate Setup)
- **RC-08:** Job Board One-Click Broadcast & Integration
- **PO-01:** Anonymized Mandate Sharing & Masking Vault

3. Step-by-Step User Journey & Execution

1. **Public Lead Capture (AS-01, AS-02):** A prospective client visits the public agency storefront (AS-01) at [https://\[subdomain\].recruiteros.com](https://[subdomain].recruiteros.com). They complete the 4-step wizard (AS-02), providing company details, role specs, budget, and selecting commercial terms (*Standard Contingency vs Priority Retainer*). The mandate is saved as **Unreviewed Inbound**.
2. **Owner Verification & Client Account Creation (AS-02 - LOCKED RULE):** The intake lands in the Cockpit. **Only Agency Owners / Team Leads** have permission to accept the inbound request. Upon acceptance, the system automatically creates a new **Client** record and dispatches an automated onboarding confirmation message via Email & WhatsApp to the client HR contact.
3. **Cockpit Mandate Assignment (RC-03):** The Owner/TL assigns the mandate to a specific Recruiter in the Cockpit. The job card initializes with strict SLA timers (e.g., Target 48 hours for initial candidate shortlist presentation).
4. **Multichannel Job Broadcast (RC-08):** The assigned Recruiter opens the job mandate, selects connected job boards (Naukri, Bayt, LinkedIn), and clicks [**Broadcast**]. The system formats payloads for each portal API and logs external Job IDs.
5. **Anonymized Partner Sharing (PO-01):** If sourcing capacity is constrained, the recruiter clicks [**Share with Partner Network**]. The system auto-masks client name (e.g., "TechCorp" becomes "Leading Tier-1 E-Commerce Platform"), strips salary/contact details, and generates an anonymized 128-bit magic link dispatched to partner sourcers.



4. Role Permissions & Access Rules

- **Inbound Mandate Conversion (AS-02):** Restricted strictly to **AGENCY_OWNER** and **TEAM_LEAD**. Standard **RECRUITER** roles can view assigned active mandates but cannot accept new client accounts or alter billing fee terms.
- **Partner Share Creation (PO-01):** Permitted for **AGENCY_OWNER**, **TEAM_LEAD**, and assigned **RECRUITER**.

5. Executable PostgreSQL Schemas

SQL

```

CREATE TABLE client_records (
  client_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  agency_id UUID NOT NULL REFERENCES agencies(agency_id) ON DELETE CASCADE,
  company_name VARCHAR(255) NOT NULL,
  primary_hr_name VARCHAR(255) NOT NULL,
  primary_hr_email VARCHAR(255) NOT NULL,
  primary_hr_phone VARCHAR(50) NOT NULL,
  agreed_fee_percentage NUMERIC(5,2) DEFAULT 8.33,
  billing_address TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

```

```

CREATE TABLE job_mandates (
  job_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  agency_id UUID NOT NULL REFERENCES agencies(agency_id) ON DELETE CASCADE,
  client_id UUID NOT NULL REFERENCES client_records(client_id) ON DELETE CASCADE,
  assigned_recruiter_id UUID REFERENCES users(user_id),
  job_title VARCHAR(255) NOT NULL,
  min_exp_years INT DEFAULT 0,

```

```

max_exp_years INT,
min_fixed_ctc NUMERIC(12,2),
max_fixed_ctc NUMERIC(12,2),
target_location VARCHAR(255) NOT NULL,
open_positions INT DEFAULT 1,
status VARCHAR(50) DEFAULT 'Active', -- 'Unreviewed Inbound', 'Active', 'On Hold', 'Closed'
created_at TIMESTAMPTZ DEFAULT NOW(),
stage_updated_at TIMESTAMPTZ DEFAULT NOW()
);

```

```

CREATE TABLE partner_mandate_shares (
  share_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  job_id UUID NOT NULL REFERENCES job_mandates(job_id) ON DELETE CASCADE,
  partner_email VARCHAR(255) NOT NULL,
  masked_job_title VARCHAR(255) NOT NULL,
  masked_company_description TEXT NOT NULL,
  partner_split_percentage NUMERIC(5,2) DEFAULT 50.00,
  access_token_hash VARCHAR(64) UNIQUE NOT NULL,
  expires_at TIMESTAMPTZ NOT NULL,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

```

6. Backend API Routes & Controller Logic

- **POST /api/v1/public/storefront/:subdomain/submit-mandate:** Public rate-limited endpoint for client self-serve intake.
- **POST /api/v1/mandates/:inbound_id/convert:** Owner-only endpoint. Converts **Unreviewed Inbound** to **Active**, creates **client_records** row, and fires onboarding email/WhatsApp webhooks.
- **POST /api/v1/jobs/:job_id/partner-share:** Generates SHA-256 token hash, regex-masks raw JD text (stripping URLs, emails, company name), and stores share record.

7. Edge Cases & Validation Rules

- **Client Duplication Guard:** When accepting an inbound mandate (**AS-02**), the controller performs an exact case-insensitive check on **company_name** and **primary_hr_email**. If existing, it links the new mandate to the existing **client_id** rather than duplicating client records.

8. UI Specification & AI Wireframe Prompt

Plaintext

Clean, modern enterprise dashboard modal. Deep Navy (#0F172A) header reads 'Convert Inbound Hiring Mandate'. Display client submission summary: Company 'Apex Cloud Labs', Role 'Sr Backend Lead', Selected Term 'Priority Retainer (5% Upfront)'. Highlighted CTA button in Botspring Yellow (#FFD400) reads 'Approve Mandate & Send Client Onboarding Package'. Restricted to Owner badge shown in top right. Generous whitespace, crisp line dividers.

WORKFLOW 2: CANDIDATE SUPPLY INGESTION, SOURCING & TALENT MAPPING

1. Workflow Business Purpose

Gathers candidate resumes from multiple channels, parses unstructured text, prevents database duplication, arbitrates partner submission rights, maps spousal/relational mobility, and recycles past vetted candidates.

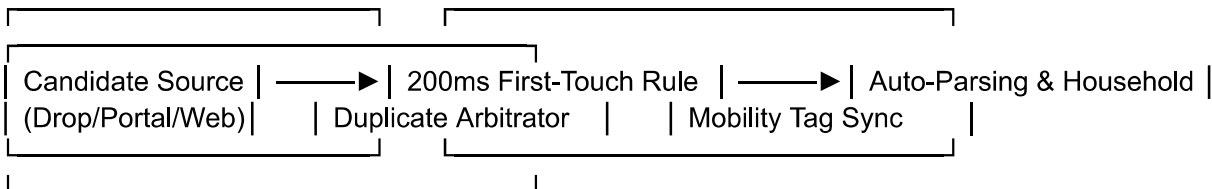
2. Integrated Feature Map

- **AS-04:** Candidate Application Portal

- **RC-02:** Automated CV Parsing & Intake Engine
- **RC-08:** Job Board Webhook Auto-Ingestion
- **PO-02:** Isolated Partner Submission Vault
- **PO-03:** Automated Candidate Ownership & Duplicate Arbitrator (200ms First-Touch Rule)
- **RC-04:** Relational Talent & Household Mapping
- **RC-07:** Silver Medalist Talent Recycler
- **AS-03:** Hot Talent Showcase & Gallery

3. Step-by-Step User Journey & Execution

1. **Multi-Channel Candidate Ingestion (AS-04, RC-02, RC-08):** Candidates enter via direct website drops (AS-04), external portal webhooks (RC-08), or manual drag-and-drop file ingestion in the Cockpit (RC-02).
2. **Partner Submission Ingestion & Duplicate Arbitration (PO-02, PO-03):** If a partner sourcer uploads a CV via their isolated vault (PO-02), the backend executes the 200ms First-Touch Arbitrator (PO-03). Phone numbers are stripped of country codes/spaces, and emails are normalized:
 - *Rule 1 (Active In-House Submission < 90 Days):* Blocked. Partner sees error modal: "Candidate active in client pipeline."
 - *Rule 2 (Prior Partner Submission):* Blocked. First-touch partner wins.
 - *Rule 3 (Unique or Stale Lead > 180 Days):* Approved. Partner awarded 50% split rights.
3. **CV Parsing & Sanitization (RC-02):** Raw files (PDF/DOCX) are parsed into structured entities (Name, Phone, Email, CTC, Notice Period, Skills). An auto-sanitized copy stripping personal contact details is generated for external presentation.
4. **Relational Household Mapping (RC-04):** For relocating candidates, recruiters link spousal or reference profiles. If Candidate A accepts a Dubai offer, linked Candidate B (Spouse) inherits tag **Geographically Mobile: Dubai (Hot Lead)**.
5. **Database Recycling & Showcase (RC-07, AS-03):** When creating a new mandate, the Cockpit highlights past 2nd/3rd place "Silver Medalists" (RC-07). Top vetted profiles are showcased anonymously on the storefront gallery (AS-03) to attract fresh employer inquiries.



4. Role Permissions & Access Rules

- **Partner Vault Access (PO-02):** Partner accounts are strictly restricted to querying **WHERE share_id = partner_token.share_id**. They can never access main agency candidate databases.
- **Internal Database Search (RC-07):** Available to all internal agency roles (**AGENCY_OWNER, TEAM_LEAD, RECRUITER**).

5. Executable PostgreSQL Schemas

SQL

```

CREATE TABLE candidate_records (
  candidate_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  agency_id UUID NOT NULL REFERENCES agencies(agency_id) ON DELETE CASCADE,
  full_name VARCHAR(255) NOT NULL,
  email VARCHAR(255) NOT NULL,
  phone_normalized VARCHAR(50) NOT NULL,
  total_exp_months INT DEFAULT 0,
  current_company VARCHAR(255),
  current_title VARCHAR(255),
  current_fixed_ctc NUMERIC(12,2),
  expected_fixed_ctc NUMERIC(12,2),

```

```

notice_period_days INT DEFAULT 30,
current_location VARCHAR(255),
primary_skills TEXT[],
raw_cv_url VARCHAR(512) NOT NULL,
sanitized_cv_url VARCHAR(512),
source_type VARCHAR(50) DEFAULT 'Direct_Upload',
last_activity_at TIMESTAMPTZ DEFAULT NOW(),
created_at TIMESTAMPTZ DEFAULT NOW(),
CONSTRAINT unique_agency_candidate_email UNIQUE (agency_id, email),
CONSTRAINT unique_agency_candidate_phone UNIQUE (agency_id, phone_normalized)
);

CREATE TABLE candidate_relational_links (
    link_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    primary_candidate_id UUID NOT NULL REFERENCES candidate_records(candidate_id) ON DELETE
    CASCADE,
    related_candidate_id UUID NOT NULL REFERENCES candidate_records(candidate_id) ON DELETE
    CASCADE,
    relationship_type VARCHAR(50) NOT NULL CHECK (relationship_type IN ('SPOUSE', 'COLLEAGUE',
    'REFERRAL')),
    inherited_target_location VARCHAR(255),
    created_at TIMESTAMPTZ DEFAULT NOW()
);

```

6. Backend API Routes & Controller Logic

- **POST /api/v1/candidates/parse-and-ingest:** Ingests multipart binary file, parses text using entity extraction, executes duplicate check, and uploads raw/sanitized copies to S3.
- **POST /api/v1/public/partner/:token/submissions:** Partner ingestion endpoint running duplicate arbitration service before DB insertion.

7. Edge Cases & Validation Rules

- **Phone Normalization Rule:** Before running duplicate checks, the API applies regex `replaceAll(/[^0-9]/g, '')` and takes the last 10 digits to neutralize formatting variances (e.g., `+91 98765-43210` vs `09876543210`).

8. UI Specification & AI Wireframe Prompt

Plaintext

Split-screen resume parsing workspace. Left side shows rendered PDF resume. Right side displays clean, structured input fields populated automatically: Full Name 'Priya Mehta', Primary Skills tags ['React', 'PostgreSQL', 'AWS'], Notice Period '30 Days'. Top banner displays 'Duplicate Check Passed - Unique Candidate Created'. Navy sidebar (#0F172A), crisp line typography, yellow action buttons (#FFD400).

WORKFLOW 3: DAILY RECRUITER COCKPIT EXECUTION & COMMUNICATION

1. Workflow Business Purpose

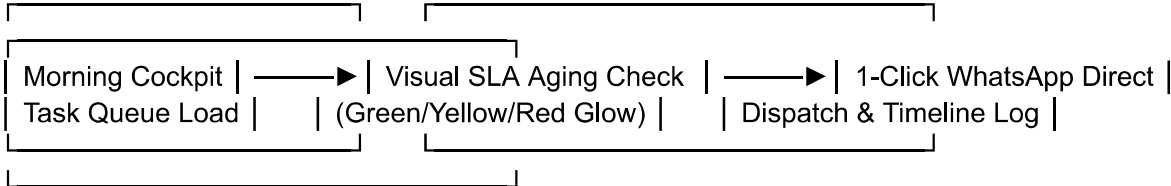
Eliminates memory-based recruiting by providing a prioritized daily action queue, visual SLA aging indicators, and unified multi-channel communication logging directly from one command panel.

2. Integrated Feature Map

- **RC-01:** Unified WhatsApp & Email Communication Log
- **RC-03:** Pipeline SLA & Stagnation Aging Radar (Daily Execution)

3. Step-by-Step User Journey & Execution

1. **Morning Workspace Load (RC-03):** Recruiter opens the Cockpit at 9:00 AM. The top panel displays their **Daily Focus Queue**, displaying tasks due today alongside cards nearing stage SLAs.
2. **Visual SLA Radar Inspection (RC-03):** Pipeline candidate cards render SLA status pills:
 - *Green:* Within SLA target (< 24 hours).
 - *Yellow:* Warning threshold reached (36 hours in stage).
 - *Subtle Red Glow:* Critical Breach (72+ hours stuck in stage without movement).
3. **Contextual Messaging & Template Dispatch (RC-01):** Clicking a candidate opens their unified timeline. The recruiter selects [**Send WhatsApp Template**] → [**Screening Follow-Up**]. The message populates variables and dispatches via WhatsApp Business API without leaving the Cockpit.
4. **Real-time Interaction Logging (RC-01):** Inbound WhatsApp replies and emails automatically append to the candidate's activity feed, resetting their **last_contacted_at** timestamp.



4. Role Permissions & Access Rules

- **Cockpit Visibility (RC-03):** Standard **RECRUITER** roles see only candidate cards and jobs assigned to them. **AGENCY_OWNER** and **TEAM_LEAD** can toggle between individual recruiter cockpits or view agency-wide aggregate boards.

5. Executable PostgreSQL Schemas

SQL

```
CREATE TABLE candidate_submissions (  
  submission_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  agency_id UUID NOT NULL REFERENCES agencies(agency_id) ON DELETE CASCADE,  
  job_id UUID NOT NULL REFERENCES job_mandates(job_id) ON DELETE CASCADE,  
  candidate_id UUID NOT NULL REFERENCES candidate_records(candidate_id) ON DELETE CASCADE,  
  stage VARCHAR(50) DEFAULT 'Screened', -- 'Screened', 'Submitted', 'Interviewing', 'Offered', 'Joined',  
  'Rejected'  
  stage_updated_at TIMESTAMPTZ DEFAULT NOW(),  
  last_communication_at TIMESTAMPTZ DEFAULT NOW(),  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

```
CREATE TABLE communication_logs (  
  log_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  agency_id UUID NOT NULL REFERENCES agencies(agency_id) ON DELETE CASCADE,  
  submission_id UUID REFERENCES candidate_submissions(submission_id) ON DELETE CASCADE,  
  channel VARCHAR(20) NOT NULL CHECK (channel IN ('WHATSAPP', 'EMAIL', 'SYSTEM_NOTE')),  
  direction VARCHAR(10) NOT NULL CHECK (direction IN ('INBOUND', 'OUTBOUND')),  
  message_body TEXT NOT NULL,  
  sent_by_user_id UUID REFERENCES users(user_id),  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

6. Backend API Routes & Controller Logic

- **GET /api/v1/cockpit/daily-queue:** Returns prioritized task list and SLA-breached candidate submissions sorted by **stage_updated_at ASC**.

- **POST /api/v1/communications/dispatch:** Sends WhatsApp Business API template/Email, logs entry to `communication_logs`, and updates `submission.last_communication_at`.

7. Edge Cases & Validation Rules

- **Negative Sentiment Keyword Detection:** If an inbound WhatsApp message contains keywords like *"declining"*, *"accepted counter offer"*, or *"not interested"*, the controller automatically triggers a **High-Risk Badge** on the candidate card in the Cockpit.

8. UI Specification & AI Wireframe Prompt

Plaintext

Recruiter daily cockpit interface. Deep Navy (#0F172A) sidebar with yellow active indicator (#FFD400). Main Kanban board shows pipeline columns: 'Screened', 'Submitted to Client', 'Interviewing'. Under 'Submitted', top card 'Neha Gupta' has subtle red border and badge 'SLA BREACH: 78h No Feedback'. Right side panel shows live embedded WhatsApp chat feed with 1-click template selection buttons.

WORKFLOW 4: CLIENT PRESENTATION, FEEDBACK & INTERVIEW SCHEDULING

1. Workflow Business Purpose

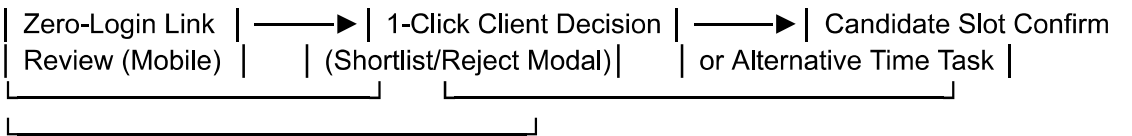
Presents candidate shortlists to hiring managers via zero-login links, captures structured rejection/shortlist decisions, handles client SLA reminders, and locks in interview calendar slots without back-and-forth ping-pong.

2. Integrated Feature Map

- **CF-01:** Zero-Login Magic Link Candidate Presenter
- **CF-02:** One-Click Candidate Decision Matrix
- **CF-04:** Automated Client Chase Sequence
- **CF-03:** Asynchronous Interview Slot Selector
- **CE-01:** 1-Click WhatsApp Slot Confirmer

3. Step-by-Step User Journey & Execution

1. **Zero-Login Shortlist Dispatch (CF-01):** The recruiter submits a candidate shortlist. The system dispatches an encrypted URL (<https://app.recruiteros.com/review/:token>) to Client HR via WhatsApp/Email.
2. **Client Portal Review (CF-01):** Client HR opens the link on mobile/desktop without a password. The page displays sanitized candidate summaries and embedded resumes (stripped of contact info).
3. **1-Click Decision & Manual Re-sourcing (CF-02 - LOCKED RULE):** The client clicks [**Shortlist**], [**Hold**], or [**Reject**]. If rejecting, a mandatory modal captures the reason (*Over Budget*, *Notice Period Too Long*, *Skill Gap*). **Rejections update candidate status to Rejected but do NOT auto-create re-sourcing tasks**, leaving re-sourcing decisions strictly to the recruiter's manual judgment.
4. **Automated Client SLA Reminders (CF-04):** If candidate links remain unreviewed after 24–48 hours, a background cron job dispatches polite WhatsApp nudges to the client.
5. **Interview Slot Selection & Alternative Slot Request (CF-03, CE-01 - LOCKED RULE):** Upon shortlisting, the client drops 3 available interview time blocks (CF-03). The options route to candidate WhatsApp (CE-01).
 - *Happy Path:* Candidate taps 1 slot $\rightarrow$ interview locked, meeting link generated, calendars synced.
 - *Alternative Path (LOCKED RULE):* If none of the 3 client slots work, the candidate clicks **"Request Alternative Slots"** in their mobile view and submits 2–3 preferred times. This immediately creates an alert task in the **assigned Recruiter's Cockpit** to coordinate with the client (preventing unvetted schedules from polluting the Client Portal).



4. Role Permissions & Access Rules

- **Client Token Scoping (CF-01):** Client access tokens expire in 14 days and are strictly scoped to `job_id`. Unauthenticated endpoints strip all candidate PII (phone, email, home address) from JSON responses.

5. Executable PostgreSQL Schemas

SQL

```
CREATE TABLE client_portal_tokens (  
  token_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  agency_id UUID NOT NULL REFERENCES agencies(agency_id) ON DELETE CASCADE,  
  job_id UUID NOT NULL REFERENCES job_mandates(job_id) ON DELETE CASCADE,  
  token_hash VARCHAR(64) UNIQUE NOT NULL,  
  expires_at TIMESTAMPTZ NOT NULL,  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);  
  
CREATE TABLE proposed_interview_slots (  
  slot_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  submission_id UUID NOT NULL REFERENCES candidate_submissions(submission_id) ON DELETE  
CASCADE,  
  interviewer_email VARCHAR(255) NOT NULL,  
  start_time TIMESTAMPTZ NOT NULL,  
  end_time TIMESTAMPTZ NOT NULL,  
  status VARCHAR(30) DEFAULT 'Proposed', -- 'Proposed', 'Accepted', 'RejectedByCandidate'  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

6. Backend API Routes & Controller Logic

- **GET /api/v1/public/portal/:token:** Public endpoint returning sanitized shortlist payload.
- **POST /api/v1/public/portal/:token/decision:** Captures shortlist/rejection decision, logs rejection reason, and resets stage SLA timer.
- **POST /api/v1/public/candidate/confirm-slot:** Confirms selected slot OR receives **Request Alternative Slots** payload, creating a Recruiter Cockpit task.

7. Edge Cases & Validation Rules

- **Past Time Guard:** The API rejects any interview slot payload where `start_time < NOW() + INTERVAL '12 hours'`.

8. UI Specification & AI Wireframe Prompt

Plaintext

Mobile web view for hiring manager candidate review. Header shows client logo 'TechCorp' and role 'Senior Backend Engineer'. Candidate card displays title 'Senior Java & AWS Specialist', experience '7.5 Years', notice '30 Days'. Floating bottom action bar has 3 high-contrast buttons: Green 'Shortlist', Dark Grey 'Hold', Red 'Reject'. Modern, zero clutter, high whitespace.

WORKFLOW 5: INTERVIEW PREPARATION, DEBRIEF & STAGE-GATE ENFORCEMENT

1. Workflow Business Purpose

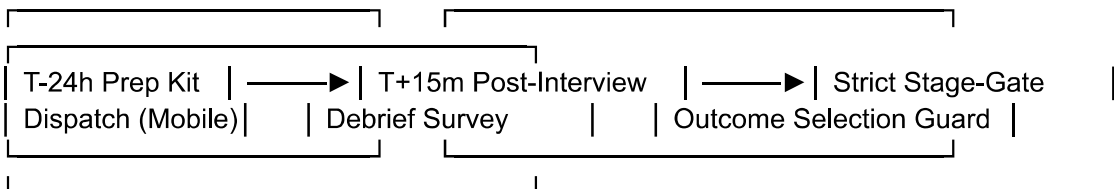
Prepares candidates for interviews using automated coaching kits, debriefs them immediately post-interview, handles no-show routing, and enforces strict stage-gate outcome logging before pipeline progression.

2. Integrated Feature Map

- **CE-02:** Automated Interview Prep Kit Trigger (T-24h)
- **CE-03:** Post-Interview Candidate Feedback Collector (T+15m)
- Stage-Gate Enforcement & No-Show Handling Engine

3. Step-by-Step User Journey & Execution

1. **Automated Prep Kit Trigger (CE-02):** Exactly 24 hours before a confirmed interview, the candidate receives an automated WhatsApp link. Opening the link renders a mobile prep card containing company intelligence, tech stack notes, interviewer details, and behavioral response frameworks (e.g., *The Orange Test*). Candidate taps [**I've Reviewed & Feel Ready!**].
2. **Post-Interview Debrief Dispatch (CE-03):** 15 minutes post-interview, the system sends a WhatsApp survey asking:
 - *Rating:* 1 to 5 Stars.
 - *Interest Level:* [**100% Excited!**], [**Have Doubts**], [**Not Interested**].
 - *Debrief Notes:* Text or 30-second audio voice note.
3. **No-Show Handling Logic (LOCKED RULE):** If the candidate or hiring manager reports a no-show, the system automatically shifts the pipeline stage to **Interview No-Show / Action Required** and creates a high-priority task in the assigned Recruiter's Cockpit to investigate and reschedule.
4. **Strict Stage-Gate Enforcement (LOCKED RULE):** The system **strictly enforces that an explicit outcome (Completed, Rescheduled, or Candidate Rejected Post-Interview) MUST be recorded** before allowing a candidate profile to advance to **Offer Issued** or **Stage Cleared**.



4. Role Permissions & Access Rules

- **Stage Progression Override:** Standard **RECRUITER** roles cannot bypass stage gates. Bypassing an unrecorded interview outcome requires explicit override authorization from **AGENCY_OWNER** or **TEAM_LEAD**.

5. Executable PostgreSQL Schemas

SQL

```
CREATE TABLE interview_schedules (  
  interview_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  submission_id UUID NOT NULL REFERENCES candidate_submissions(submission_id) ON DELETE  
CASCADE,  
  confirmed_slot_id UUID NOT NULL REFERENCES proposed_interview_slots(slot_id),  
  meeting_link VARCHAR(512),  
  outcome_status VARCHAR(50) DEFAULT 'Scheduled', -- 'Scheduled', 'Completed', 'Rescheduled', 'No_Show',  
'Rejected_Post_Interview'  
  candidate_prep_acknowledged BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMPTZ DEFAULT NOW(),  
  updated_at TIMESTAMPTZ DEFAULT NOW()  
);
```

```
CREATE TABLE interview_debriefs (
  debrief_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  interview_id UUID NOT NULL REFERENCES interview_schedules(interview_id) ON DELETE CASCADE,
  rating INT CHECK (rating >= 1 AND rating <= 5),
  interest_level VARCHAR(50) NOT NULL, -- '100% Excited', 'Have Doubts', 'Not Interested'
  candidate_notes TEXT,
  voice_note_url VARCHAR(512),
  submitted_at TIMESTAMPTZ DEFAULT NOW()
);
```

6. Backend API Routes & Controller Logic

- **GET /api/v1/public/candidate/prep-kit/:token:** Renders candidate preparation data and logs view timestamp.
- **POST /api/v1/public/candidate/debrief:** Ingests rating, text, or voice note payload and appends to candidate timeline.
- **PATCH /api/v1/submissions/:submission_id/stage:** Stage progression route that validates `interview_schedules.outcome_status != 'Scheduled'` before allowing transition to **Offered**.

7. Edge Cases & Validation Rules

- **Unprepped Alert:** If **T-4 Hours** before interview and `candidate_prep_acknowledged == FALSE`, dispatch an urgent WhatsApp reminder to candidate and trigger an alert icon in the Cockpit.

8. UI Specification & AI Wireframe Prompt

Plaintext

Mobile post-interview debrief screen. Header reads 'TechCorp — Interview Debrief'. 5-star rating selector component at top. 3 large choice cards for Interest Level: '100% Excited to Join!' (Highlighted in Botspring Yellow #FFD400), 'Have Doubts', 'Not Interested'. Audio record button labeled 'Tap to record 30-sec voice debrief'. Dark navy button 'Submit Debrief'.

WORKFLOW 6: OFFER AUDIT, COMPLIANCE & NOTICE-PERIOD RETENTION

1. Workflow Business Purpose

Audits commercial offer terms, verifies placement fees, monitors candidate notice-period retention risks using 2-tier escalation rules, and collects pre-onboarding compliance documentation.

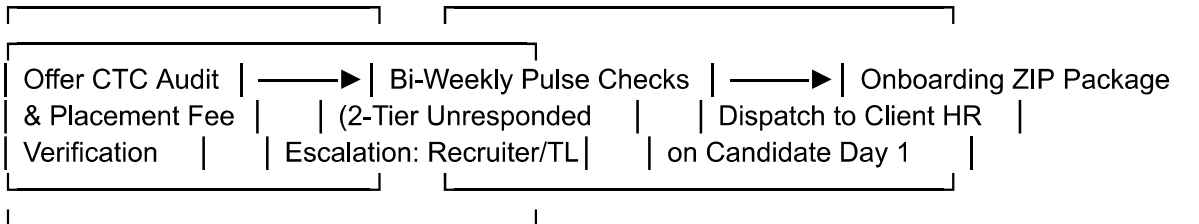
2. Integrated Feature Map

- **HC-02:** Offer Audit & CTC Verification Engine
- **RC-05:** Post-Offer 90-Day Drop-Off Radar
- **CE-04:** Notice Period Counter-Offer Pulse Checks
- **HC-01:** Automated Compliance Document Vault
- **HC-03:** Zero-Touch Client HR Handoff Portal

3. Step-by-Step User Journey & Execution

1. **Offer CTC Verification & Auto-Invoicing Pre-Draft (HC-02):** Upon offer acceptance, the recruiter inputs signed fixed/variable CTC terms and joining date. The system auto-calculates placement fee (**Fixed CTC** $\times$ **Agreed Fee %**), checks CTC expectation variances, and pre-drafts the upcoming invoice.
2. **Bi-Weekly Notice-Period Pulse Checks (RC-05, CE-04):** Every 14 days during the candidate's 30/60/90-day notice period, an automated WhatsApp pulse check dispatches to the candidate's phone asking about resignation acceptance, handover progress, and counter-offers.
3. **2-Tier Unresponded Pulse Check Escalation (LOCKED RULE):**

- **1st Unresponded Attempt (48 Hours):** Candidate risk score automatically shifts to **HIGH (Red Warning Banner)** in the assigned Recruiter's Daily Cockpit with an urgent task: *"Candidate dark on Day [X] pulse check — Call immediately"*.
 - **2nd Consecutive Unresponded Attempt:** System dispatches an **automated escalation alert directly to the Agency Owner / Team Lead** to intervene and protect revenue.
4. **Mobile Compliance Vault Upload (HC-01):** The candidate receives a mobile checklist to upload compliance files (National ID, Pay Slips, Degree Certificates, Relieving Letter).
 5. **Zero-Touch HR Package Handoff (HC-03):** On Day 1, Client HR receives a secure link to download the complete onboarding ZIP package and confirm physical joining with one click.



4. Role Permissions & Access Rules

- **Offer Financial Approval (HC-02):** Financial fields on offer audits (CTC modifications, fee percentage overrides) can be edited by recruiters during draft, but final approval locks editing rights to **AGENCY_OWNER** or **TEAM_LEAD**.

5. Executable PostgreSQL Schemas

SQL

```

CREATE TABLE job_offer_audits (
  audit_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  submission_id UUID NOT NULL REFERENCES candidate_submissions(submission_id) ON DELETE
  CASCADE,
  offered_fixed_ctc NUMERIC(12,2) NOT NULL,
  offered_variable_ctc NUMERIC(12,2) DEFAULT 0.00,
  agreed_fee_percentage NUMERIC(5,2) NOT NULL,
  calculated_placement_fee NUMERIC(12,2) NOT NULL,
  joining_date DATE NOT NULL,
  signed_offer_url VARCHAR(512) NOT NULL,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
  
```

```

CREATE TABLE notice_period_pulse_logs (
  pulse_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  submission_id UUID NOT NULL REFERENCES candidate_submissions(submission_id) ON DELETE
  CASCADE,
  touchpoint_day INT NOT NULL,
  response_status VARCHAR(30) DEFAULT 'Pending', -- 'Responded_Safe', 'High_Risk_Counter_Offer',
  'Unresponded'
  unresponded_attempts INT DEFAULT 0,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
  
```

6. Backend API Routes & Controller Logic

- **POST /api/v1/offers/audit:** Saves offer CTC metrics, verifies fee calculation, and pre-drafts invoice.
- **POST /api/v1/public/candidate/notice-pulse:** Candidate pulse check submission endpoint. Calculates risk score and updates **notice_period_pulse_logs**.

- **Cron Worker (2-Tier Escalation):** Daily cron job checks for unresponded pulses > 48h. Attempt 1 triggers Recruiter Cockpit task. Attempt 2 triggers Owner/TL escalation notification.

7. Edge Cases & Validation Rules

- **Private PII S3 Signed URLs:** Pre-onboarding compliance files (Pay Slips, National IDs) stored in S3 must be accessed strictly through temporary signed URLs with expiration `expiresIn = 900s`.

8. UI Specification & AI Wireframe Prompt

Plaintext
Notice period risk radar view in recruiter cockpit. Horizontal timeline bar shows candidate 'Vikram Malhotra — Day 35 of 60'. Day 15 milestone shows green checkmark 'Resignation Confirmed'. Day 35 milestone displays pulsating red alert badge 'HIGH RISK: Unresponded to 2nd Pulse Check Attempt'. Escalation notice box reads 'Escalated to Team Lead'. Yellow call action button (#FFD400).

WORKFLOW 7: SETTLEMENT, PROBATION BREACH & PARTNER LEDGERS

1. Workflow Business Purpose

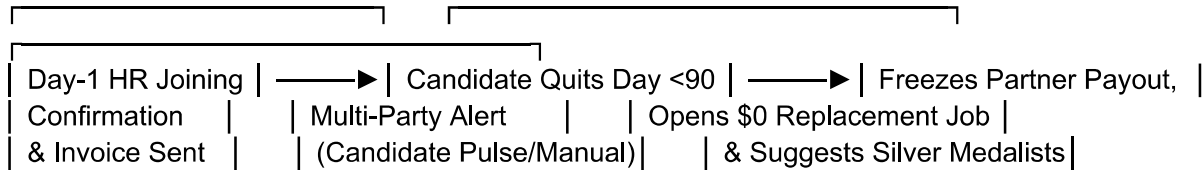
Dispatches invoices upon Day 1 candidate joining, monitors 90-day probation replacement guarantees, executes multi-party early departure alerts, and manages split-commission operational ledgers.

2. Integrated Feature Map

- **RC-06:** Lifecycle-Triggered Settlement Engine
- **HC-04:** Probation Guarantee Clock & Milestone Tracker
- **PO-04:** Split-Fee Ledger & Auto-Settlement Interlink

3. Step-by-Step User Journey & Execution

1. **Day-1 Invoice Dispatch & Probation Clock Activation (RC-06, HC-04):** When Client HR confirms candidate physical joining in **HC-03**, the pre-drafted placement invoice dispatches to Client Finance, and a 90-day probation replacement guarantee clock activates in the Cockpit (**HC-04**).
2. **Early Departure / Probation Breach Workflow (HC-04, RC-06, PO-04 - LOCKED RULES):**
 - *Trigger:* Candidate indicates early departure (< 90 days) via WhatsApp pulse check (**CE-04**), or client/recruiter manually logs departure.
 - *Multi-Party Alerts (LOCKED RULE):* System immediately dispatches high-priority alerts to **Agency Owner / Team Lead, Primary Assigned Recruiter, and Partner Recruiter** (if a 50% split partner exists).
 - *Automated Replacement Operations:* System **automatically freezes any pending Partner Split Commission payouts (PO-04)**, opens a \$0 "Free Replacement Mandate" linked to the client, and **auto-suggests a shortlist of past Silver Medalist candidates** for instant re-interviewing.
3. **Owner Financial Approval Enforcement (RC-06 - LOCKED RULE):** Issuing Credit Notes, refunds, or invoice cancellations **strictly requires explicit approval from the Agency Owner / Team Lead** (Recruiters cannot alter or cancel financial invoices).
4. **Manual Ledger Management (PO-04 - LOCKED RULE):** Automated payment gateway payouts are strictly out of scope. The system operates purely as an operational ledger where payment statuses (*Invoice Issued, Client Paid, Partner Payout Authorized*) are marked manually by the Agency Owner based on real-world cash collections.



4. Role Permissions & Access Rules

- **Financial Adjustments (RC-06):** Approving Credit Notes, fee refunds, or marking Partner Payables as **Paid** is restricted strictly to **AGENCY_OWNER** and **TEAM_LEAD**.

5. Executable PostgreSQL Schemas

SQL

```
CREATE TABLE invoice_records (  
  invoice_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  agency_id UUID NOT NULL REFERENCES agencies(agency_id) ON DELETE CASCADE,  
  client_id UUID NOT NULL REFERENCES client_records(client_id),  
  submission_id UUID NOT NULL REFERENCES candidate_submissions(submission_id),  
  invoice_number VARCHAR(100) UNIQUE NOT NULL,  
  amount NUMERIC(12,2) NOT NULL,  
  status VARCHAR(50) DEFAULT 'Draft', -- 'Draft', 'Sent', 'Paid', 'Credit_Note_Issued', 'Cancelled'  
  issued_at TIMESTAMPTZ,  
  paid_at TIMESTAMPTZ,  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

```
CREATE TABLE probation_guarantee_trackers (  
  guarantee_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  submission_id UUID NOT NULL REFERENCES candidate_submissions(submission_id) ON DELETE  
CASCADE,  
  joining_date DATE NOT NULL,  
  expiry_date DATE NOT NULL,  
  status VARCHAR(50) DEFAULT 'Active_Probation', -- 'Active_Probation', 'Completed', 'Breached_Quitted'  
  replacement_mandate_id UUID REFERENCES job_mandates(job_id),  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

```
CREATE TABLE partner_split_ledgers (  
  ledger_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  submission_id UUID NOT NULL REFERENCES candidate_submissions(submission_id) ON DELETE  
CASCADE,  
  share_id UUID NOT NULL REFERENCES partner_mandate_shares(share_id),  
  partner_share_amount NUMERIC(12,2) NOT NULL,  
  payout_status VARCHAR(50) DEFAULT 'Pending_Client_Payment', -- 'Pending_Client_Payment',  
'Frozen_Probation_Breach', 'Ready_For_Payout', 'Paid'  
  updated_by_user_id UUID REFERENCES users(user_id),  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

6. Backend API Routes & Controller Logic

- **POST /api/v1/probation/breach:** Triggered upon early candidate departure (<90 days). Sets **guarantee.status = 'Breached_Quitted'**, updates **partner_split_ledgers.payout_status = 'Frozen_Probation_Breach'**, creates \$0 replacement mandate, and dispatches multi-party alerts.
- **POST /api/v1/invoices/:invoice_id/credit-note:** Owner-only endpoint. Validates **user.role IN ('AGENCY_OWNER', 'TEAM_LEAD')** before generating a credit note record.

7. Edge Cases & Validation Rules

- **Financial Permission Guard:** Any HTTP **POST/PATCH** attempt by a **RECRUITER** role to alter **invoice_records** or **partner_split_ledgers** must be blocked at middleware with HTTP **403 Forbidden**.

8. UI Specification & AI Wireframe Prompt

Plaintext
Financial settlements and probation tracker dashboard. Summary cards at top: 'Unbilled Placements', 'Active 90-Day Guarantees', 'Frozen Split Payouts'. Table row displays 'Apex Corp — Candidate: Ankit Sharma', Status 'PROBATION BREACH (Quitted Day 42)'. Purple badge reads 'Partner Payout FROZEN'. Prominent yellow CTA button (#FFD400) reads 'View Suggested Silver Medalists for \$0 Replacement Mandate'. Restricted Owner actions banner. Deep Navy theme.

CONSOLIDATED MASTER FEATURE COVERAGE MATRIX

Feature Code	Feature Name	Primary Assigned Workflow
FOUNDATION-01	Multi-Tenant Architecture & RLS	System Infrastructure (Underpins Workflows 1–7)
AS-01	Public Agency Storefront	Workflow 1 (Demand Acquisition)
AS-02	Self-Serve Client Mandate Ingestion	Workflow 1 (Demand Acquisition)
AS-03	Hot Talent Showcase Gallery	Workflow 2 (Supply Ingestion)
AS-04	Candidate Application Portal	Workflow 2 (Supply Ingestion)
RC-01	Unified WhatsApp & Email Log	Workflow 3 (Daily Execution)
RC-02	Auto CV Parsing & Intake Engine	Workflow 2 (Supply Ingestion)
RC-03	Pipeline SLA & Stagnation Aging Radar	Workflow 1, Workflow 3

RC-04	Relational Talent & Household Mapping	Workflow 2 (Supply Ingestion)
RC-05	Post-Offer 90-Day Drop-Off Radar	Workflow 6 (Offer & Notice Period)
RC-06	Lifecycle-Triggered Settlement Engine	Workflow 7 (Settlement & Probation)
RC-07	Talent Database Recycling Engine	Workflow 2 (Supply Ingestion)
RC-08	Job Board Multi-Broadcast & Webhooks	Workflow 1, Workflow 2
CF-01	Zero-Login Magic Link Candidate Presenter	Workflow 4 (Client Presentation)
CF-02	One-Click Candidate Decision Matrix	Workflow 4 (Client Presentation)
CF-03	Asynchronous Interview Slot Selector	Workflow 4 (Client Presentation)
CF-04	Automated Client Chase Sequence	Workflow 4 (Client Presentation)
CE-01	1-Click WhatsApp Slot Confirmator	Workflow 4 (Client Presentation)
CE-02	Automated Interview Prep Kit Trigger	Workflow 5 (Interview Prep & Debrief)
CE-03	Post-Interview Candidate Feedback Collector	Workflow 5 (Interview Prep & Debrief)

CE-04	Notice Period Counter-Offer Pulse Checks	Workflow 6 (Offer & Notice Period)
HC-01	Automated Compliance Document Vault	Workflow 6 (Offer & Notice Period)
HC-02	Offer Audit & CTC Verification Engine	Workflow 6 (Offer & Notice Period)
HC-03	Zero-Touch Client HR Handoff Portal	Workflow 6, Workflow 7
HC-04	Probation Guarantee Clock & Milestone Tracker	Workflow 7 (Settlement & Probation)
PO-01	Anonymized Mandate Sharing & Masking Vault	Workflow 1 (Demand Acquisition)
PO-02	Isolated Partner Submission Vault	Workflow 2 (Supply Ingestion)
PO-03	Automated Candidate Ownership Arbitrator	Workflow 2 (Supply Ingestion)
PO-04	Split-Fee Ledger & Auto-Settlement Interlink	Workflow 7 (Settlement & Probation)